

Tutorial Sheet 1 (Compressible Flow)

1. Air at a velocity of 210 ms^{-1} decelerates through a diffuser to a velocity of 60 ms^{-1} . The temperature and pressure at the inlet are 278 K and 80 kPa and the exit pressure is 90 kPa . Calculate
 - (a) the change in stagnation pressure, and
 - (b) the change in entropy.

[Ans: -12.1 kPa , $35.22 \text{ J kg}^{-1} \text{ K}^{-1}$]

(a) Assume a calorically perfect gas

$$M_1 = \frac{u_1}{a_1} = \frac{u_1}{\sqrt{\gamma R T_1}} = 0.6283$$

$$p_{01} = \frac{p_{01}}{p_1} p_1 = \left(1 + \frac{\gamma-1}{2} M_1^2\right)^{\frac{\gamma}{\gamma-1}} p_1 = 104.3703 \text{ kPa}$$

Assume flow is steady, total enthalpy H is constant along a streamline

$$\therefore C_p T_1 + \frac{u_1^2}{2} = C_p T_2 + \frac{u_2^2}{2} \quad \text{where } C_p = \frac{\gamma R}{\gamma-1} = 1004.5 \text{ J/kgK}$$

$$\therefore T_2 = 298.1593 \text{ K}$$

$$\therefore M_2 = \frac{u_2}{a_2} = 0.1733$$

$$\therefore p_{02} = \left(1 + \frac{\gamma-1}{2} M_2^2\right)^{\frac{\gamma}{\gamma-1}} p_2 = 91.9074 \text{ kPa}$$

$$\therefore \Delta p_0 = p_{02} - p_{01} = -12.4709 \text{ kPa}$$

$$\begin{aligned} \text{(b) } \Delta S &= C_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} \\ &= 36.52 \text{ J/kgK} \end{aligned}$$

2. Air enters a constant-area duct with the conditions shown in figure 1. Heat is added to the flow at a rate $q = 3 \times 10^5 \text{ J kg}^{-1}$.

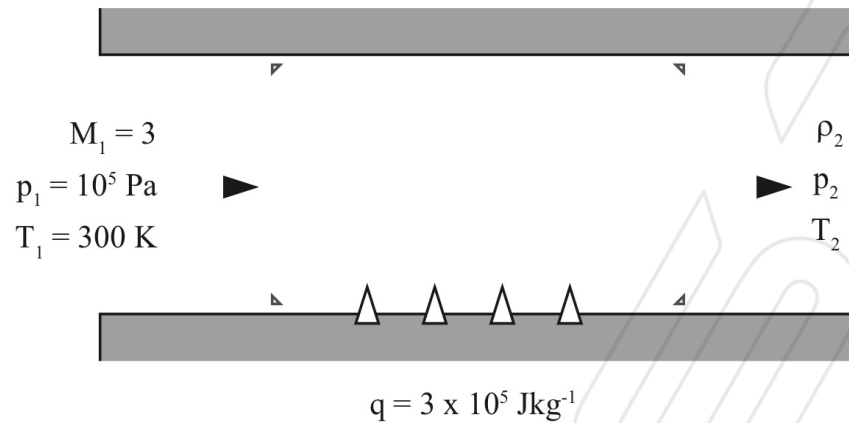


Figure 1

Find the Mach number, static pressure and temperature, stagnation pressure and temperature at the exit of the duct.

[Ans: $T_{02} = 1139 \text{ K}$, $M_2 = 1.59$, $p_2 = 299 \text{ kPa}$, $T_2 = 761 \text{ K}$, $p_{02} = 1250 \text{ kPa}$]

$$T_{01} = \frac{T_{01}}{T_1} T_1 = \left(1 + \frac{\gamma-1}{2} M_1^2\right) T_1 = 840 \text{ K}$$

$$T_{02} = T_{01} + \frac{q}{C_p} = T_{01} + \frac{q}{\frac{\gamma}{\gamma-1} R} = 1138.66 \text{ K}$$

$$\therefore \frac{T_{02}}{T_{01}} = \frac{T_{02}}{T_{01}} \cdot \frac{T_{01}}{T_{01}} = 0.65398 \cdot \frac{1138.66 \text{ K}}{840 \text{ K}} = 0.88650$$

$$\therefore M_2 = 1.59$$

$$\therefore T_2 = \frac{T_2}{T_{02}} \left(\frac{T_{01}}{T_{02}}\right)^{-1} T_1 = 756.43 \text{ K}$$

$$p_2 = \frac{p_2}{p_{02}} \left(\frac{p_{01}}{p_{02}}\right)^{-1} p_1 = 2.9962 \times 10^5 \text{ Pa}$$

$$\rho_2 = \frac{p_2}{R T_2} = 1.3801 \text{ kg/m}^3$$

$$\therefore p_{02} = \left(1 + \frac{\gamma-1}{2} M_2^2\right)^{\frac{\gamma}{\gamma-1}} p_2 = 1.2548 \times 10^6 \text{ Pa}$$

3. A symmetrical two-dimensional airfoil with thickness-to-chord ratio 0.07 is placed in figure 2 is placed at an angle of attack of 7° in a supersonic stream of air at Mach 2.5. Compute the lift and drag coefficients.

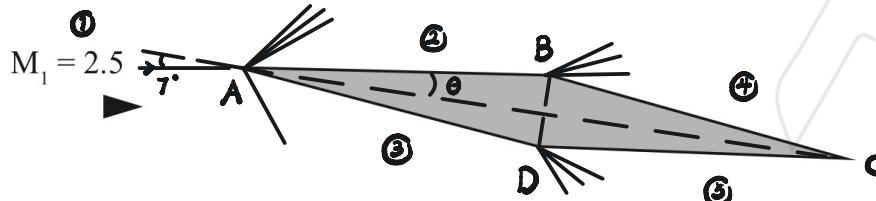


Figure 2

[Ans: $c_l = 0.2163$ and $c_d = 0.0362$]

$$\frac{t}{c} = 0.07$$

$$\alpha = 7^\circ = \frac{7\pi}{180} \text{ rad}$$

$$\therefore \theta = \tan^{-1}(0.07) = 4.0042^\circ$$

As shown in annotated Figure 2,

for ②

$$\theta_1 - \gamma(M_1) = \theta_2 - \gamma(M_2)$$

$$\therefore 0^\circ - 39.12^\circ = (7^\circ - 4.0042^\circ) - \gamma(M_2) \quad \therefore \gamma(M_2) = 42.1158^\circ \quad \therefore M_2 = 2.63$$

$$\therefore \frac{p_2}{p_1} = \frac{p_2}{p_{2s}} \frac{p_{2s}}{p_1} = \frac{17.0859}{20.9015} = 0.8174$$

for ③

$$\delta = 7^\circ + 4.0042^\circ = 11.0042^\circ \quad \therefore \theta = 32.841^\circ$$

$$\therefore M_3 = 2.042$$

$$\frac{p_3}{p_1} = 1.9795$$

Therefore,

for ④

$$\theta_1 - \gamma(M_1) = \theta_4 - \gamma(M_4) \quad \therefore 0^\circ - 42.1158^\circ = 2(4.0042^\circ) - \gamma(M_4)$$

$$\therefore \gamma(M_4) = 50.1242^\circ \quad \therefore M_4 = 3.02$$

$$\therefore \frac{p_4}{p_1} = \frac{p_4}{p_{4s}} \frac{p_{4s}}{p_1} = \frac{p_4}{p_{04}} \frac{p_{0s}}{p_1} = \frac{17.0859}{37.8503} = 0.4514$$

for ⑤

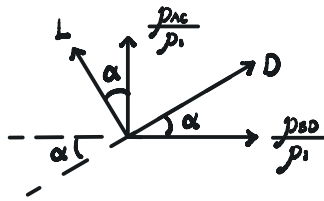
$$\nu(M_5) = \nu(M_3) + 2(4.0042)^\circ = 27.5308^\circ + 8.0084^\circ = 35.5392^\circ$$

$$\therefore M_5 = 2.35$$

$$\therefore \frac{p_5}{p_1} = \frac{p_5}{p_{os}} \frac{p_{os}}{p_s} \frac{p_s}{p_1} = \frac{8.3533}{13.5214} \cdot 1.9795 = 1.2229$$

$$\therefore \frac{p_{Ac}}{p_1} (\uparrow) = \left(\frac{p_s}{p_1} + \frac{p_s}{p_1} - \frac{p_u}{p_1} - \frac{p_v}{p_1} \right) \cos \theta = 1.9289$$

$$\frac{p_{so}}{p_1} (\rightarrow) = \left(\frac{p_u}{p_1} + \frac{p_s}{p_1} - \frac{p_u}{p_1} - \frac{p_s}{p_1} \right) \sin \theta = 0.07839$$



$$\therefore \frac{p_L}{p_1} = \frac{p_{Ac}}{p_1} \cos \alpha - \frac{p_{so}}{p_1} \sin \alpha = 1.9050$$

$$\frac{p_o}{p_1} = \frac{p_{Ac}}{p_1} \sin \alpha + \frac{p_{so}}{p_1} \cos \alpha = 0.3129$$

In 2D.

$$C_L = \frac{L}{\frac{1}{2} \rho_\infty U_\infty^2 c} = \frac{2 p_L c}{\rho_1 (M_1^2 \delta R T_1) c} = 2 \frac{p_L}{p_1} \frac{1}{\delta M_1^2} = 0.4354$$

$$C_D = \frac{D}{\frac{1}{2} \rho_\infty U_\infty^2 c} = 2 \frac{p_o}{p_1} \frac{1}{\delta M_1^2} = 0.07152$$

for F = force per unit width.

$$\frac{F_{Ac}}{p_1} (\uparrow) = \left(\frac{p_s}{p_1} + \frac{p_s}{p_1} - \frac{p_u}{p_1} - \frac{p_v}{p_1} \right) \cos \theta \cdot \frac{c}{2} \frac{1}{\cos \theta} = 0.9668 c$$

$$\frac{F_{so}}{p_1} (\rightarrow) = \left(\frac{p_u}{p_1} + \frac{p_s}{p_1} - \frac{p_u}{p_1} - \frac{p_s}{p_1} \right) \sin \theta \cdot \frac{c}{2} \frac{1}{\cos \theta} = 0.03929 c$$

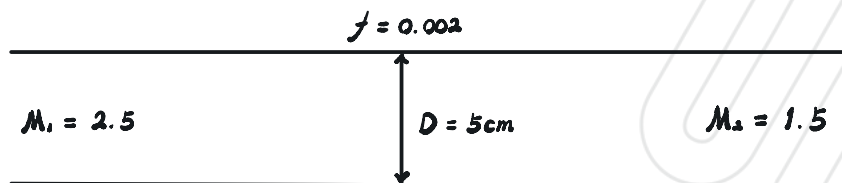
$$\therefore \frac{F_L}{p_1} = \frac{F_{Ac}}{p_1} \cos \alpha - \frac{F_{so}}{p_1} \sin \alpha = 0.9548 c$$

$$\frac{F_o}{p_1} = \frac{F_{Ac}}{p_1} \sin \alpha + \frac{F_{so}}{p_1} \cos \alpha = 0.1568 c$$

$$\therefore C_L = \frac{L}{\frac{1}{2} \rho_\infty U_\infty^2 c} = 2 \frac{F_L}{p_1} \frac{1}{\delta M_1^2 c} = 0.2182, C_D = 2 \frac{F_o}{p_1} \frac{1}{\delta M_1^2 c} = 0.03584$$

4. Air flows adiabatically in a 5 cm inner diameter circular duct. The air decelerates from Mach 2.5 at section 1 to Mach 1.5 at section 2. What is the length of the duct between two the sections? What is the maximum length of the duct if no shock occurs? Assume $f = \text{constant} = 0.002$.

[Ans: $L = 1.85 \text{ m}$, $L_{\max} = 2.7 \text{ m}$]



$$\left(\frac{4fL}{D}\right)_2 = \left(\frac{4fL}{D}\right)_1 - \frac{4fL}{D}$$

$$\therefore \frac{4fL}{D} = 0.4320 - 0.1361 = 0.2959$$

$$\therefore L = 184.9375 \text{ cm} = 1.849 \text{ m}$$

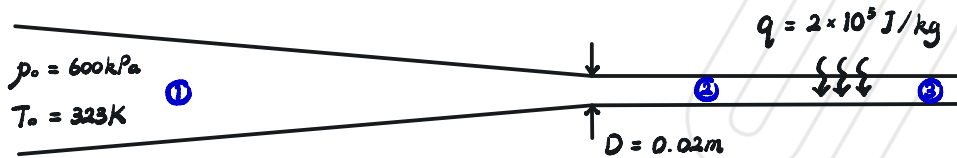
For no shock occurs, $M_2 \rightarrow 1$

$$\frac{4fL_{\max}}{D} = 0.4320 - 0 = 0.4320$$

$$\therefore L_{\max} = 270 \text{ cm} = 2.7 \text{ m}$$

5. Air at a stagnation pressure of 600 kPa and stagnation temperature of 323 K flows through an isentropic convergent nozzle. The nozzle feeds a 0.02 m diameter duct. Heat is transferred in to this duct from an external source at 200 kJ kg^{-1} of air. Determine the maximum mass flow rate and the range of back pressure for maximum rate of flow.

[Ans: 0.29 kg s^{-1} , Pressure range: 0 – 279 kPa]



$$T_{02} = T_0 + \frac{q}{C_p} = T_0 + \frac{\gamma-1}{\gamma R} q = 522.10 \text{ K}$$

for isentropic nozzle,

$$p_{02} = p_0 = 600 \text{ kPa}$$

$$\dot{m} = \rho u A = \rho M \sqrt{\gamma R T} A = \frac{p}{R T} M \sqrt{\gamma R T} A = p \sqrt{\frac{\gamma}{R T}} M \pi \left(\frac{D}{2}\right)^2$$

$$= p_{02} \left(\frac{p_{02}}{p}\right)^{1/\gamma} \sqrt{\frac{\gamma}{R T_{02}}} \left(\frac{T_{02}}{T}\right)^{1/2} M \pi \left(\frac{D}{2}\right)^2$$

$$= p_{02} \sqrt{\frac{\gamma}{R T_{02}}} \left(\frac{T_{02}}{T}\right)^{-\frac{\gamma}{\gamma-1} + \frac{1}{2}} M \pi \left(\frac{D}{2}\right)^2$$

$$= p_{02} \sqrt{\frac{\gamma}{R T_{02}}} \frac{\pi D^2}{4} M \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma-1}{2(\gamma-1)}}$$

$$\therefore \frac{d\dot{m}}{dM} = p_{02} \sqrt{\frac{\gamma}{R T_{02}}} \frac{\pi D^2}{4} \left[\left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma-1}{2(\gamma-1)}} + M \frac{\gamma-1}{2(\gamma-1)} (\gamma-1) M \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{1-3\gamma}{2(\gamma-1)}} \right]$$

$$= p_{02} \sqrt{\frac{\gamma}{R T_{02}}} \frac{\pi D^2}{4} \left[\left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma-1}{2(\gamma-1)}} - \frac{\gamma+1}{2} M^2 \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{1-3\gamma}{2(\gamma-1)}} \right]$$

for $\frac{d\dot{m}}{dM} = 0$, $\dot{m}_{max}$ occurs

$$\therefore \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma-1}{2(\gamma-1)}} - \frac{\gamma+1}{2} M^2 \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{1-3\gamma}{2(\gamma-1)}} = 0$$

$$\therefore \frac{\gamma+1}{2} M^2 = 1 + \frac{\gamma-1}{2} M^2$$

$$\therefore M^2 = 1$$

$$\therefore M = 1$$

$$\therefore \dot{m}_{max} = 0.333 \text{ kg/s}$$

For maximum M before heat transferred.

$$M_3 = 1 \text{ (choked)}$$

$$\therefore \frac{T_{03}}{T_0^*} = 1$$

$$\therefore \frac{T_{03}}{T_0^*} = \frac{T_{03}}{T_{03}} \frac{T_{03}}{T_0^*} = 0.6187$$

$$\therefore M_2 = 0.45$$

$$\therefore \dot{m}_{\max} = 0.2926 \text{ kg/s}$$

$$\text{for } M_2 = 0.45, M_3 = 1$$

$$(p_{\text{back}})_{\min} = 0 \text{ (choked)}$$

$$(p_{\text{back}})_{\max} = (p_2)_{\max} \frac{p_3}{p_2} = \left(\frac{p_0}{p_2}\right)_{\max}^{-1} p_0 \frac{p_3}{p_0} \left(\frac{p_3}{p_0}\right)^{-1}$$

$$= \left(1 + \frac{\gamma-1}{2} M_2^2\right)^{-\frac{\gamma}{\gamma-1}} p_0 \frac{1}{1.86485}$$

$$= 0.4654 p_0$$

$$= 279.25 \text{ kPa}$$

$$\therefore p_{\text{back}} = 0 \sim 279.25 \text{ kPa}$$

6. Consider a 15° half-angle wedge at zero angle of attack in a Mach 3 airflow as shown in figure 3.

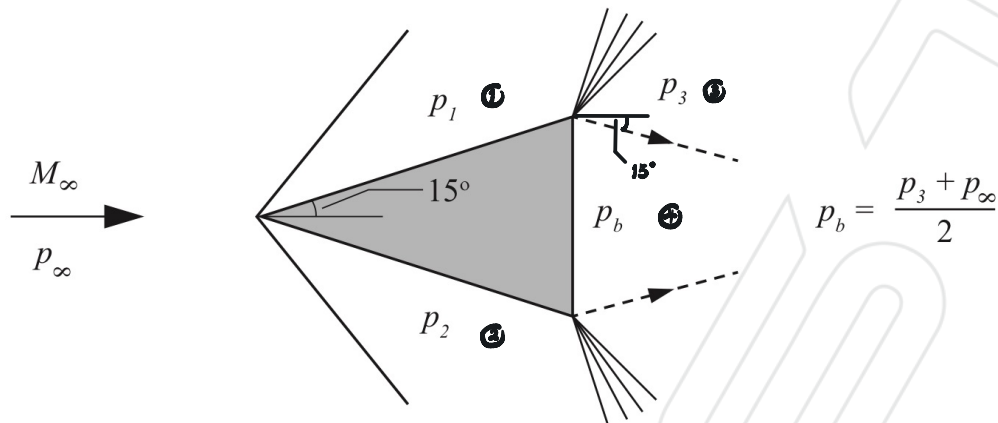


Figure 3

Assuming that flow separation behind the wedge causes the streamline at the trailing edge to be deflected towards the base by 15° from the horizontal and that the base pressure is the average of the free-stream pressure and the pressure in region 3, Calculate the lift and drag coefficients of this wedge.

[Ans: $c_l = 0$ and $c_d = 0.186$]

At $\alpha = 0^\circ$

for ② and ③

$$M_\infty = 3, \theta = 15^\circ$$

$$\therefore M_1 = M_2 = 2.252$$

$$\frac{p_1}{p_\infty} = \frac{p_2}{p_\infty} = 2.8305$$

for ③

$$M_1 = 2.252$$

$$\therefore \theta_1 - \gamma(M_1) = \theta_3 - \gamma(M_3)$$

$$\therefore -15^\circ - 33.0704^\circ = 15^\circ - \gamma(M_3)$$

$$\therefore \gamma(M_3) = 63.0704^\circ$$

$$\therefore M_3 = 3.80$$

$$\frac{p_\infty}{p_3} = 115.89$$

$$\therefore \frac{p_3}{p_\infty} = \left(\frac{p_\infty}{p_3}\right)^{-1} \frac{p_1}{p_1} \frac{p_1}{p_\infty} = \frac{11.599}{115.89} \cdot 2.8305 = 0.2833$$

$$\therefore p_b = \frac{p_3 + p_\infty}{2} = 0.64165 p_\infty$$

$$\therefore F_1 = p_0 \frac{c}{\cos 15^\circ} = 2.9303 p_0 c$$

$$F_2 = 2.9303 p_0 c$$

$$F_3 = p_0 \cdot 2(c \tan 15^\circ) = 0.34386 p_0 c$$

$$\therefore L = (F_1 - F_2) \cos 15^\circ = 0$$

$$D = (F_1 + F_2) \sin 15^\circ - F_3 = 1.1730 p_0 c$$

$$\therefore C_L = \frac{L}{\frac{1}{2} \rho_\infty U_\infty^2 c} = 0$$

$$C_D = \frac{D}{\frac{1}{2} \rho_\infty U_\infty^2 c} = \frac{1.1730 p_0 c}{\frac{1}{2} \rho_\infty M_\infty^2 \delta R T_\infty c} = \frac{2 \cdot 1.1730 p_0}{\rho_\infty M_\infty^2 \delta} = \frac{2.3459}{M_\infty^2 \delta} = 0.1862$$